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Optical spectroscopy study of neodymium in sodium alumino-borosilicate glasses

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Abstract

Over the composition range from peraluminous to peralkaline ($R_{PA} = Na_2O/(Na_2O + Al_2O_3)$ from 0 to 1), ultraviolet–visible spectroscopy was used to study the local environment of Nd–O in glasses (molar composition): $60SiO_2 \cdot 15B_2O_3 \cdot xNa_2O \cdot (25 - x)Al_2O_3 \cdot yNd_2O_3$ ($0 \leqslant x \leqslant 25$ and y = solubility limit). Our study focused on Judd–Ofelt oscillator strength parameters: Ω_2 – ligand field asymmetry and Ω_6 – bond covalence. We explain the similarity between Ω_2/Ω_6 and the Nd_2O_3 solubility curve in terms of Nd partitioning into borate- and silicate-rich environments. However, Ω_2/Ω_6 is nearly independent of R_{PA} for the glasses containing Nd_2O_3 at the solubility limit irrespective of glass composition, suggesting that average local Nd–O environment is nearly the same for all glasses.

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1. Introduction

Recently, our studies have shown similar solution behaviors and solubility limits for La_2O_3, Nd_2O_3, and Gd_2O_3, in sodium alumino-borosilicate glasses, in which the peralkali ratio ($R_{PA} = Na_2O/(Na_2O + Al_2O_3)$) varied from 0 to 1 [1–3]. Separately we studied Ln-induced phase separation using La_2O_3 and Gd_2O_3 and found that glasses prone to phase separation over $0.3 < R_{PA} < 0.65$ contained a dispersed amorphous phase enriched in B and La or Gd [4,5]. On either side of this range, no phase separation was observed for the glasses containing Ln(III) (Ln = La, Gd, or Nd), but of the silicate $NaLn_{9.33-x/3}Si_6O_{26}$ (x = 0 to 1), depending on R_{PA}, crystallized above the Ln solubility limit. As an example (Fig. 1), when $R_{PA} < 0.4$ mullite ($Al_6Si_2O_{13}$) formed from the melt, the extent of which decreases with increasing Nd concentration and completely dissolved when Nd reaches a certain concentration, depending on R_{PA}. Above the Nd solubility limit, $Na_xNd_{9.33-x/3}Si_6O_{26}$ (x = 0 to 1) precipitated from the melts. For $R_{PA} > 0.4$, it is easier to form a glass independent of Ln_2O_3.

Selectively, we studied local structure of boron in the Gd-containing glasses ($R_{PA} = 0.8$) with extended electron energy loss fine structure (EXELFS) [6] and structure evolution in Gd-containing glasses ($R_{PA} = 0.8$) with Raman spectroscopy [7]. From these studies, we concluded that Ln cations generally and preferentially partition to borate-rich environment in sodium alumino-borosilicate glasses if the Ln concentration was below a threshold Ln:(B + Al) = 1:3. The threshold is governed by the local Ln environment that resembles Ln-metaborate, $2BO_3$:1Ln:$1BO_4$, in which tetrahedral AlO_4 substitutes for tetrahedral boron, BO_4.

Over the glass composition region where $0.4 \leqslant R_{PA} \leqslant 0.6$, Ln-induced phase separation was further investigated with electron transmission microscopy

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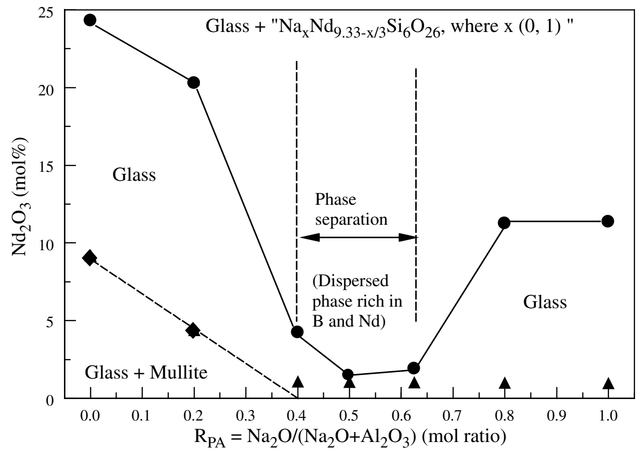

Fig. 1. Solubility limits of $60SiO_2 \cdot 15B_2O_3 \cdot xNa_2O \cdot (25-x)Al_2O_3 \cdot yNd_2O_3$ ($0 = x = 25$; y = solubility limit) as a function of R_{PA} from 0 to 1 (the solubility limits were determined at temperature of 1450 °C using quenched glasses as detailed elsewhere [2,3]; the limits have an accuracy better than 0.1 mol% for 0.1 mass% increase in Nd_2O_3 in the glass results in precipitation of the Nd-silicate phase. The dispersed phase in the phase separated glasses is believed to be enriched in B and Nd based on previous TEM study of La- or Gd-containing glasses having the same compositions [6]).

(TEM) [8]. We found that the glasses with little excess Al_2O_3 or excess Na_2O, where the excess was determined by $|Al_2O_3–Na_2O|$, were prone to Ln-induced phase separation, the extent of which depended on the Ln_2O_3 to B_2O_3 ratio governed by Ln: 1/3 (B + Al). Using our proposed Ln-metaborate model, we hypothesized that once the borate sites were saturated with Ln cations, excess Ln cations dissolve in the silicate-rich environment from which the Ln-silicate eventually crystallized.

The Judd–Ofelt (J–O) theory [9,10] has been widely used to study the local Ln environment in glasses [11–15]. In the J–O theory, Ω_2 primarily represents the asymmetry of the Ln–O ligand field in the host matrix, Ω_6 primarily represents the degree of Ln–O covalency, and Ω_4 primarily represents the 'long-range' structural interaction on the local Ln environment. Examples of the 'long-range' structural interations are viscosity at high temperature and ionic packing density at low temperature of a host matrix [11–13]. Gatterer [14] and Gatterer et al. [15] used UV–VIS spectroscopy to study the sodium borate, sodium silicate, and sodium borate silicates at a constant Nd_2O_3 concentration, for which partitioning of Nd in both environments was elucidated. We previously used the same method to study the change in the local Nd–O environment as a function of Nd_2O_3 concentration in alumino-borosilicate glasses ($R_{PA} = 0$) and used the Nd partitioning mechanism to explain the observed changes in local Nd–O environment [16]. Here we systematically studied the local Nd–O ligand field asymmetry and bond covalency, JO parameters Ω_λ (where $\lambda = 2, 4$, and 6), over the composition range of R_{PA} from 0 to 1 and two Nd_2O_3 concentrations. The objective of the current study was to investigate any link between local Nd–O environment, ligand field asymmetry and bond covalency, and Nd solubility limit. In addition, the broad range of compositions covered in this study allowed us to further test our hypothesis of Ln-metal borate local environment in these melts by using glass samples that were quenched from the melt.

2. Experimental

Multicomponent borosilicate glasses, $60SiO_2 \cdot 15B_2O_3 \cdot xNa_2O \cdot (25-x)Al_2O_3 \cdot yNd_2O_3$ ($0 \leqslant x \leqslant 25$ and y = solubility limit) were prepared in 10 g batches. Each batch was prepared from reagent grade chemicals and was heated at 1550 °C for one hour and 1450 °C for another hour in sealed Pt crucibles [2]. Table 1 summarizes the glass target compositions and R_{PA}. The sample ID, for example R02-Nd5, represents the glass R_{PA} value

Table 1
Glass target compositions

Glass ID	Oxide, mol%					R_{PA} (mole ratio)
	SiO_2	B_2O_3	Na_2O	Al_2O_3	Nd_2O_3	$Na_2O/(Na_2 + Al_2O_3)$
R0-Nd32	54.51	13.63	0.00	22.71	9.15	0.00
R0-Nd60	45.42	11.35	0.00	18.92	24.30	0.00
R02-Nd18	57.38	14.34	4.78	19.13	4.37	0.20
R02-Nd55	47.83	11.96	3.99	15.94	20.28	0.20
R04-Nd5	59.37	14.84	9.89	14.84	1.05	0.40
R04-Nd18	57.45	14.36	9.58	14.36	4.25	0.40
R05-Nd5	59.38	14.84	12.37	12.37	1.04	0.50
R06-Nd5	59.39	14.85	15.47	9.28	1.02	0.63
R06-Nd9	58.86	14.72	15.33	9.20	1.90	0.63
R08-Nd5	59.41	14.85	19.80	4.95	0.99	0.80
R08-Nd40	53.25	13.31	17.75	4.44	11.26	0.80
R1-Nd5	59.42	14.86	24.76	0.00	0.96	1.00
R1-Nd41	53.19	13.30	22.16	0.00	11.35	1.00

Table 2
Glasses density, calculated Nd concentration, and refractive index function

Glass ID	Nd_2O_3 (mol%)	C_{Nd} (cations/cm^3)	$R_{Nd–Nd}$ (nm)[a]	Density (g/cm^3)	Refractive index $n = a_0 + a_1\sigma + a_2\sigma^2$ (σ in eV)		
					a_0	a_1	a_2
R0-Nd32	9.15	3.39×10^{21}	0.41	2.958	1.58449	0.00025	0.00411
R0-Nd60	24.30	8.27×10^{21}	0.31	3.851	1.72559	−0.00698	0.00603
R02-Nd18	4.37	1.72×10^{21}	0.52	2.664	1.53620	−0.00460	0.00501
R02-Nd55	20.28	7.25×10^{21}	0.32	3.685	1.67596	−0.00393	0.00552
R04-Nd5	1.05	4.42×10^{20}	0.81	2.472	1.49010	0.00645	0.00291
R04-Nd18	4.25	1.73×10^{21}	0.52	2.684	1.52150	0.00060	0.00400
R05-Nd5	1.04	4.44×10^{20}	0.81	2.482	1.48682	0.00666	0.00308
R06-Nd5	1.02	4.48×10^{20}	0.81	2.502	1.49788	−0.00382	0.00468
R06-Nd9	1.90	8.24×10^{20}	0.66	2.559	1.51450	−0.00065	0.00393
R08-Nd5	0.99	4.51×10^{20}	0.81	2.522	1.52662	−0.00893	0.00548
R08-Nd40	11.26	4.68×10^{21}	0.37	3.267	1.58524	0.00260	0.00361
R1-Nd5	0.96	4.55×10^{20}	0.81	2.542	1.50980	−0.00160	0.00382
R1-Nd41	11.35	4.87×10^{21}	0.37	3.317	1.60829	−0.00665	0.00453

[a] Assuming homogeneous distribution of Nd cations in the homogeneous host matrix, the distance between two adjacent cations is approximated as $(4\pi/3C_{Nd})^{-1/3}$.

(0.2) and Nd_2O_3 concentration (5 mass%). The target glass compositions are close to the true composition because less than 1% mass loss was measured for each glass after melting [2]. A separate study on similar glass compositions also showed that the measured and target compositions were the same within analytical uncertainty [17].

A Cary 500 UV–VIS-NIR (Varian) spectrophotometer was used to measure optical absorption spectra of the glasses over the wavelength from 400 to 900 nm. Refractive index values (n) of the glasses were measured as a function of photon energy from 1.2 eV (1033 nm) to 3.8 eV (326 nm) with a step of 0.05 eV with a UVISEL spectroscopic phase modulated ellipsometer. Refractive index functions (cf. Table 2) were derived from a second order polynomial least square fit to the measured data over the photon energy range and the regression coefficients (r^2) were better than 0.98 for all glasses. Glass densities were measured with a helium pycnometer (AccuPyc 1330, Micromiretics Corp); each reported density is an average of ten measurements with accuracy better than 0.1%.

3. Results

Densities for the Nd-containing glasses are summarized in Table 2, which were used to calculate the Nd concentrations (cation/cm^3) (Table 2). Ultraviolet–visible absorption spectra, normalized to the sample thickness, for the glasses at two Nd_2O_3 concentrations are shown in Fig. 2. The intensities of the absorption bands increase with increasing Nd_2O_3. Neodymium f-f transition bands were assigned according to Carnall et al. [18]. The overall band of $^4I_{9/2} \rightarrow {}^4G_{5/2}, {}^4G_{7/2}$ transitions, at 584 and 526 nm, respectively, is sensitive to the Nd–O ligand field asymmetry, known as the hypersensitive structure transition (HST) band. As illustrated in Fig. 2(a–e), as R_{PA} increases from 0 to 1, $^4I_{9/2} \rightarrow {}^4G_{7/2}$ of the HST band increases and the shape of the HST band becomes less symmetric, especially for the R1-Nd5 with 1 mol% Nd_2O_3 and $R_{PA} = 1$. The shapes of the other bands appeared to remain unchanged.

4. Optical oscillator strength of Nd – Judd–Ofelt theory

The absorption bands between 400 and 900 nm are characteristic of the Nd^{3+} $4f^3$ transitions from the ground state manifold, $^4I_{9/2}$, to a number of excited states that are crystal-field split, except for the $^4I_{9/2} \rightarrow {}^2P_{1/2}$ that is the terminal state or a degenerate Kramers doublet and not split by crystal-fields [19]. The crystal-field splitting results from asymmetry of Nd–O local structure, which is affected by the host matrix or ligand field around Nd cations as well as by its own concentration. A lack of long-range order in glass results in a distribution of local Nd–O environments that result in inhomogeneous broadening of the $4f^3$ transitions. At room temperature thermal broadening is considered to be small for all glasses [20,21].

From the UV–VIS spectra (Fig. 2), the electric-dipole oscillator strengths, P_{ex} can be determined from [18]

$$P_{ex} = 4.32 \times 10^{-9} \frac{9n}{(n^2+2)^2} \int \varepsilon(\sigma)\,d\sigma, \tag{1}$$

where n is the refractive index, ε is the molar absorptivity (in l/mol-cm) of a band at a mean energy σ (cm^{-1}). The oscillator strength of the same electrical dipole transitions, P_{ED}, by Judd [9] and Ofelt [10] theory, can be expressed as

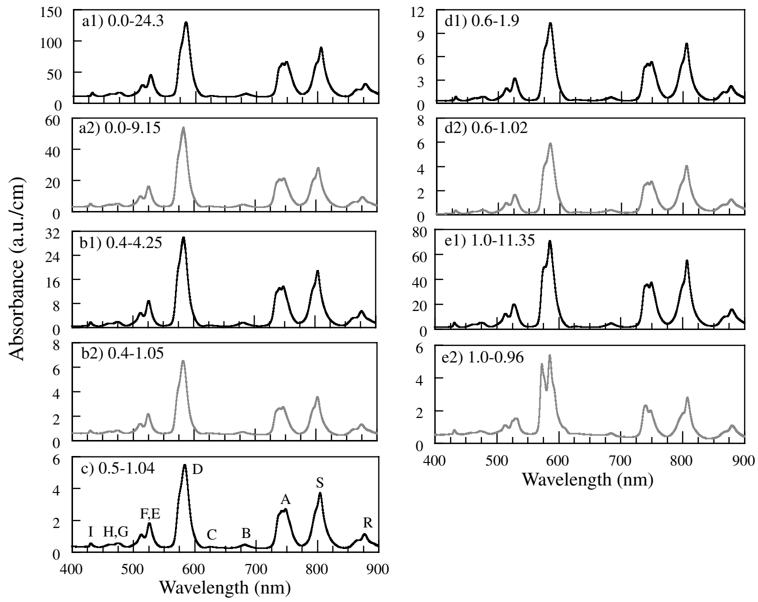

Fig. 2. Normalized UV–VIS spectra by specimen thickness for $60SiO_2 \cdot 15B_2O_3 \cdot xNa_2O \cdot (25-x)Al_2O_3 \cdot yNd_2O_3$ with different R_{PA} (left number) and Nd_2O_3 (right number) concentrations (each band is assigned according to literature [18] as R – $^4F_{3/2}$, S – $^4F_{5/2}$, $^2H_{9/2}$, A – $^4F_{7/2}$,$^4S_{3/2}$, B – $^4F_{9/2}$, C – $^2F_{11/2}$, D – $^4G_{5/2}$, $^2G_{7/2}$, E – $^4G_{7/2}$, F – $^4G_{9/2}$, G–H – $^2K_{15/2}$,$^4G_{11/2}$, I – $^2P_{1/2}$).

$$P_{ED} = \frac{8\pi^2 mc\sigma}{3h(2J+1)} \sum_{\lambda=2,4,6} \Omega_\lambda < SLJ\|U^{(\lambda)}\|S'L'J'>^2, \quad (2)$$

where m is the mass of the electron, c is the speed of light, h is the Plank's constant, Ω_λ ($\lambda = 2,4,6$) is the oscillator parameter, $\langle SLJ\|U^{(\lambda)}\|S'L'J'\rangle^2$ are the doubly reduced unit tensor operators that were calculated by Carnall et al. [18] for each transition from the initial state (SLJ) to the final state ($S'L'J'$) where S, L, J are the spin, orbital, and total angular momentum quantum numbers, respectively, of the $^{2S+1}L_J$ levels involved in the transition, and J is the total angular momentum of the initial level ($J = 9/2$). The Ω_λ can be readily determined from a linear least squares regression method fitting P_{ex} to P_{ED} for each transition. The goodness of the fit (gof) is given by the reduced chi-square defined as

$$\text{gof} = \left(\frac{\sum(P_{ED} - P_{ex})^2}{\text{number of observations} - \text{number of parameters}}\right)^{1/2}. \quad (3)$$

The details of determine P_{ex} and J–O parameters have been reported elsewhere [16]. Table 3 summarizes Ω_λ, one standard deviation, goodness of fit, and the experimental and calculated oscillator strengths, P_{ex} and P_{ED}.

5. Discussion

The host matrix affects the local Nd–O environment and in turn the HST band shape [14]. As shown in

Table 3
Judd–Oflet oscillator strength parameters ($\Omega_2, \Omega_4, \Omega_6$), experimental and modeling derived total oscillator strength (P_{ex}, P_{ED}), and goodness of the fit (gof)

Glass ID	Ω_2 (10^{-20} cm^2)	Ω_4 (10^{-20} cm^2)	Ω_6 (10^{-20} cm^2)	P_{ex} (10^{-5})	P_{ED} (10^{-5})	gof (10^{-5})
R0-Nd32	5.63 ± 0.32	3.79 ± 0.38	4.20 ± 0.27	3.58	3.17	0.19
R0-Nd60	4.53 ± 0.27	4.12 ± 0.33	4.69 ± 0.23	3.58	3.14	0.21
R02-Nd18	5.62 ± 0.30	3.73 ± 0.36	4.25 ± 0.25	3.57	3.15	0.20
R02-Nd55	4.46 ± 0.26	4.29 ± 0.32	4.80 ± 0.23	3.66	3.21	0.22
R04-Nd5	5.12 ± 0.31	3.39 ± 0.38	4.01 ± 0.28	3.28	2.89	0.18
R04-Nd18	5.62 ± 0.30	3.73 ± 0.36	4.25 ± 0.25	3.57	3.15	0.20
R05-Nd5	4.63 ± 0.29	3.54 ± 0.35	4.55 ± 0.25	3.36	2.92	0.20
R06-Nd5	5.43 ± 0.30	4.18 ± 0.36	4.59 ± 0.25	3.75	3.32	0.20
R06-Nd9	4.55 ± 0.28	4.21 ± 0.34	4.70 ± 0.24	3.62	3.18	0.21
R08-Nd5	4.25 ± 0.20	2.62 ± 0.24	2.53 ± 0.17	2.38	2.20	0.10
R08-Nd40	3.67 ± 0.28	3.64 ± 0.34	4.12 ± 0.24	3.10	2.70	0.18
R1-Nd5	4.46 ± 0.12	3.40 ± 0.14	2.77 ± 0.10	2.81	2.57	0.12
R1-Nd41	4.87 ± 0.30	4.77 ± 0.36	5.17 ± 0.26	4.00	3.52	0.22

Fig. 2, we found the HST band shapes resembled those found in Nd-containing borate glasses [14], i.e., more or less symmetric, for glasses with $R_{PA} \leqslant 0.5$. However, the band split became more pronounced as R_{PA} increased from 0.6 to 1.0, especially at 1 mol% Nd_2O_3; the band shape more resembles those found for Nd-containing silicate glasses [14]. It follows that more Nd dissolves or partitions into the borate-rich sites than into the silicate-rich sites in the Na_2O deficient host matrices. The effect of Na_2O on Nd partitioning has been previously reported [15].

As host matrix R_{PA} varies from 0 to 1, partitioning of Nd_2O_3 is expected to change Ω_λ. In principle, high asymmetry of the ligand field increases Ω_2 and high-bond covalence decreases Ω_6 [22,23]. The J–O parameter Ω_4 is also related to bond covalence from longer-range network effect [11–13]. In this study, Ω_2 generally decreases with increasing Nd_2O_3 concentration, i.e., the ligand field around Nd becomes more symmetric and independent of R_{PA}. At the same time, Nd–O bond covalence (or $1/\Omega_6$) appears to decrease, suggesting that the Nd–O bond becomes more ionic. Ionic bonding is less directional than covalent bonding, which is in good agreement with the decrease in the bond asymmetry or decrease in Ω_2. Furthermore, the increase in Ω_4 with increasing Nd_2O_3 concentration indicates possible longer-range host matrix effects on the local Nd–O environment, such as an increase in the glass packing density [11,12]. In turn, the increase in Ω_4 agrees with the fact that as more Nd cations dissolve in the host glass, the distance between adjacent Nd^{3+} cations decreases (Table 2), which likely increases the repulsion force between two neighboring Nd cations and, hence, increases Ω_4. Another possibility is the repulsive forces that could exist between Nd cations and other adjacent glass modifying cations, such as Na^+ if present in the glass. Being network-modifying cations, Nd^{3+} and Na^+ do not share the same non-bridging oxygen [24]. It follows that as Nd increases the repulsion force on Nd cations from neighboring Na cations, if present, could also contribute to the increase in Ω_4.

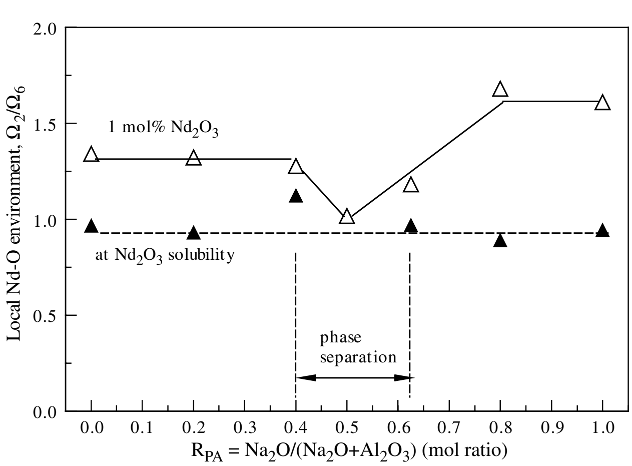

Fig. 3. Local Nd–O environment, Ω_2/Ω_6, as a function of R_{PA} showing a resemblance to the Nd-solubility curve for the low-Nd glasses.

Changes in the overall local environment for Nd can be also characterized by Ω_2/Ω_6 as a function of R_{PA} as shown in Fig. 3. At 1 mol% Nd_2O_3, for peraluminous glasses with R_{PA} between 0 and 0.4, the Ω_2/Ω_6 is nearly constant. The same is true for peralkaline glasses with R_{PA} between 0.8 and 1.0. These results suggest little change in Nd–O local environment(s) within each R_{PA} region, yet their local environments could be different. For the low-Nd glasses, the Ω_2/Ω_6 trend is qualitatively parallel to the Nd solubility curve (Fig. 1) and the local minimum of Ω_2/Ω_6 interestingly coincides with the minimum of the Nd-solubility. Furthermore, the region where the Ω_2/Ω_6 decreases coincides with the glasses prone to phase separation.

For the high-Nd glasses, independent of R_{PA}, Ω_2/Ω_6 is lower than the values for low-Nd glasses, which can result from a decrease in ligand field asymmetry, a decrease in bond covalency or both. As discussed next, lower Ω_2/Ω_6 in high-Nd glasses should relate to the last case. Although there is no direct evidence to quantify these factors, Nd in glass more or less functions as a network modifier [7,24,25] and increasing Nd creates more non-bridging oxygens. As a result, a decrease in Nd–O bond covalency is expected. Secondly, as the solubility limit for Nd is approached, more ordered Nd–O local structures, in both borate-rich and silicate-rich sites, are expected.

6. Conclusions

Combining UV–VIS spectroscopy with the J–O theory, local environments of Nd–O were systematically and qualitatively examined for sodium alumino-borosilicate glasses varying R_{PA} from 0 to 1. Composition affects partitioning of Nd_2O_3 in borate-rich and silicate-rich environments as indicated by the characteristic variations of the HST ($^4I_{9/2} \rightarrow {}^4G_{5/2}, {}^4G_{7/2}$) band shape. For baseline compositions with $R_{PA} \leqslant 0.5$, HST band shape more resembles Nd in the borate environment suggesting that more Nd partitioned to the borate sites. For baseline glasses with $R_{PA} \geqslant 0.6$, the band shape more resembles Nd in the silicate environment suggesting that more Nd partitioned to the silicate sites. Over the entire region, a similar trend between Ω_2/Ω_6 at low-Nd concentration and the solubility of Nd was found. However, Ω_2/Ω_6 is nearly constant independent of R_{PA} for glasses containing Nd_2O_3 at its solubility limit. Although Nd-saturation limits are significantly different for different glass compositions, the constant Ω_2/Ω_6 suggests that the average local Nd–O environment is nearly the same for all glasses.

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